

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

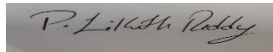
Annexure A

Analytical Problem Solving

Code of Conduct:

I P LIKITH REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

ANSWER:

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation is used to **enhance image quality** by modifying the intensity values of pixels. It helps improve **contrast**, highlight important details, and make objects in an image more visible.

(b) Negative Transformation

Formula:

$$s = 255 - r$$

Given:

$$r = [50, 100, 150, 200]$$

Calculations:

- $255 - 50 = 205$
- $255 - 100 = 155$
- $255 - 150 = 105$
- $255 - 200 = 55$

Answer:

$$s = [205, 155, 105, 55]$$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c\log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

ANSWER:

(a) Purpose

Log transformation **expands low-intensity (dark) pixel values** and compresses high-intensity values. This improves the visibility of details in dark regions of an image.

(b) Calculation

Formula:

$$s = c\log(1 + r)$$

Given:

- $c = 10$

- $r = [0, 10, 50, 100]$
(Using natural logarithm, $\ln$)

r	Calculations	s
0	$10\ln(1)$	0.00
10	$10\ln(11)$	23.98
50	$10\ln(51)$	39.32
100	$10\ln(101)$	46.15

Answer:

$$s = [0, 23.98, 39.32, 46.15]$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

ANSWER:

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization improves **image contrast** by redistributing pixel intensity values over the available range, making dark and bright regions more distinguishable.

(b) Cumulative Distribution and Equalization

Given histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels:

$$N = 2 + 2 + 4 + 8 = 16$$

Step 1: Cumulative Distribution

Intensity	Frequency	Cumulative Frequency
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Intensities

For a 3-bit image:

$$L = 8$$

Formula:

$$s_k = \frac{\text{CDF}}{N} (L - 1)$$

where $L - 1 = 7$.

Intensity	CDF	Equalized Value
0	2	$\frac{2}{16} \times 7 = 0.875 \approx 1$
1	4	$\frac{4}{16} \times 7 = 1.75 \approx 2$
2	8	$\frac{8}{16} \times 7 = 3.5 \approx 4$
3	16	$\frac{16}{16} \times 7 = 7$

Answer:

Original	Equalized
0	1
1	2
2	4

Original	Equalized
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

ANSWER:

4. Spatial Filtering (Smoothing)

(a) Purpose

Spatial smoothing removes **noise**, reduces small intensity variations, and produces a smoother image by averaging neighboring pixels.

(b) Apply 3×3 Averaging Filter

Given matrix:

10	20	30
20	30	40
30	40	50

Sum of all values:

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average:

$$\frac{270}{9} = 30$$

New center pixel value:

30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

[0 -1 0], {-1 4 -1}, {0 -1 0}

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

ANSWER:

5. Spatial Filtering (Sharpening)

(a) Purpose

Sharpening enhances **edges and fine details** by emphasizing rapid changes in pixel intensity, making the image appear clearer.

(b) Apply Laplacian Mask

Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Center output:

$$\begin{aligned} & (4 \times 20) - (10 + 10 + 10 + 10) \\ &= 80 - 40 \\ &= 40 \end{aligned}$$

Answer:

$$\boxed{40}$$